IB MYP DESIGN • CRITERIA A

Inquiring and Analysing

Design Research Workbook • Investigating problems, exploring existing solutions, and developing design briefs.



THE MISSION

Welcome to your Design Workbook

This workspace guides you step-by-step through Criterion A. You will act as an empathetic investigator, analyzing real-world systems to build a robust launchpad for your design solutions.

WORKBOOK OBJECTIVES

In this workbook, you will:

🔍**Investigate & Justify:** Identify real problems and analyze their impact on people, the planet, and prosperity.

📚**Evaluate Evidence:** Collect, catalog, and evaluate research information from highly credible and reliable sources.

🔎**Analyze Solutions:** Deconstruct existing products using key criteria such as Purpose, Context, Parts, and Complexity.

📝**Synthesize & Brief:** Combine your research findings to draft a clear, fully justified, and actionable design brief.

Global Problem Identification & Analysis Workspace

My Problem Statement

[Insert your clearly defined problem statement here. What is the specific issue you are aiming to solve?]

Context of the Problem

Guiding Prompts:

- Where does this occur?
- Who experiences it?
- Under what circumstances?

[Insert context of the problem here...]

Evidence the Problem Exists

[Insert statistics, key scientific facts, data points, or peer-reviewed literature citations proving this problem exists.]

Why is this Globally Important?

[Connect this local/specific issue to the UN Sustainable Development Goals (SDGs). Why does this challenge matter on a global scale?]

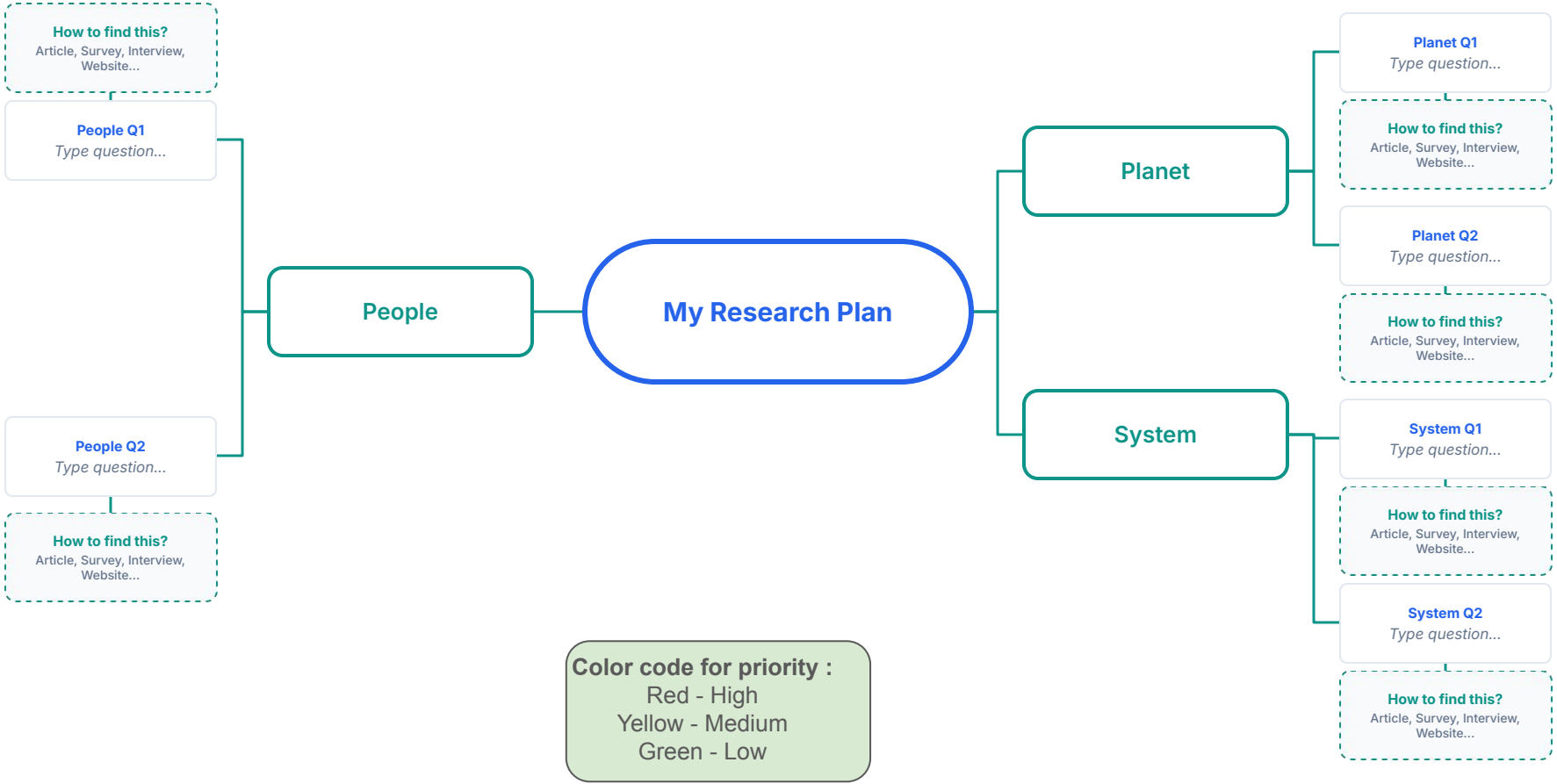
Why Does This Problem Need a Solution?

[Describe the negative consequences of leaving this problem unsolved, and outline the positive outcomes of designing an effective solution. Who will benefit most from this development?]

Effects of the Problem

Effect	Who is affected?	Evidence
[e.g., Habitat loss]	[e.g., Wildlife/Biota]	[e.g., 40% decline]
[Describe effect 2]	[Identify target group]	[Provide evidence]

Research Plan



Existing Product Analysis & Evaluation

Identify design features, purposes, and user contexts to form insights for your design criteria

Product	Parts	Purpose	Complexity
PRODUCT 01 [Name / Model]	[Describe key parts and components...]	[Describe primary purpose and target user...]	[Describe design and manufacturing complexity...]
PRODUCT 02 [Name / Model]	[Describe key parts and components...]	[Describe primary purpose and target user...]	[Describe design and manufacturing complexity...]
PRODUCT 03 [Name / Model]	[Describe key parts and components...]	[Describe primary purpose and target user...]	[Describe design and manufacturing complexity...]

Design Inspiration Resources: | Cooper Hewitt Design Museum • IDEO Design Kit

Research Synthesis

After analysing the existing products, I will adopt _____ because... I will adapt _____ by... I will avoid _____ because... Therefore, my design will...

Complete the scaffolding prompts above to transition from analysis to action.

SYNTHESIS & FINAL PRODUCT DIRECTION

Use this space to draft your comprehensive design synthesis. Expand on your choices above—explain how the combination of adopted features, adapted mechanisms, and avoided pitfalls will culminate in a cohesive, innovative solution that directly satisfies your design brief.

Authentic Design Resources: **Design Council UK** - https://www.designcouncil.org.uk | **Behance** - https://www.behance.net

Design Brief & Planning

Synthesizing research insights to define your product design directions

Research Findings

What were the most important findings from your research?

Existing Product Analysis

What did you learn from analysing existing products?

Proposed Solution Concept

Briefly describe the overall idea of the product you plan to design

Design Brief Starter

Based on my research and analysis, I intend to design a product that will help _____ by _____. The product is intended for _____ and aims to solve _____ while considering _____.

My Structured Design Brief

Type your final, structured design brief here based on the sentence starter above...

RESOURCES: Stanford d.school (dschool.stanford.edu) Interaction Design Foundation (interaction-design.org)

DEVELOPING DESIGN IDEAS

Criteria B – Developing Ideas

Sensing Our World: Engineering Solutions for Community Wellbeing

Strand i & ii • Specifications & Design Ideas

Strand i: Design Specifications must be synthesized into precise, measurable criteria to guide your community-focused solution.

Strand ii: Develop a range of **three feasible design ideas** and annotate them thoroughly to explain structural and functional decisions.

Strand iii • Decision & Justification

Decision Matrix: Quantitatively evaluate your three design concepts against established specifications to choose the optimal solution.

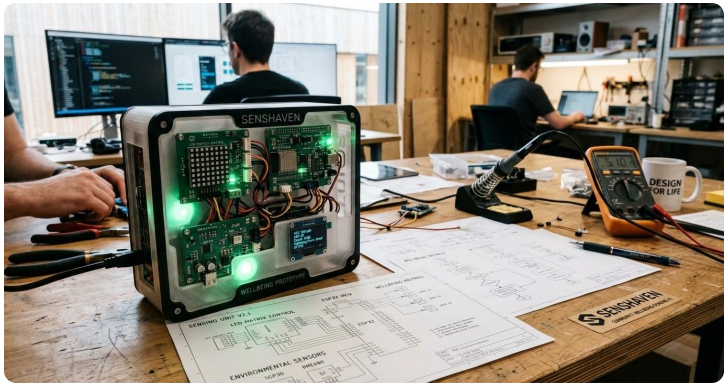
Justification: Write a robust, evidence-based justification defending your final selected design before beginning prototype creation.

Strand iv • System Architecture & Flow Diagrams

System Architecture Diagram: Illustrate physical layout, hardware modules, community touchpoints, and functional links.

System Flow Diagram: Map logical operations, sensor thresholds, input-output feedback loops, and sequential process flows.

Ensure your physical prototype matches the precise routing mapped in these high-level diagrams.



Objective Requirements & Scope

Prior to physical prototyping, students must develop, communicate, and justify an engineering design solution that directly addresses a real-world community problem. All work must be meticulously documented across the four critical strands above.

Submission Checklist: ✔ All screenshots are clear & readable • ✔ Each section is complete • ✔ Selected design matches planning diagrams

Criteria B Strand i – Design Specifications

Synthesize findings into precise, measurable criteria to guide your design solution

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Design Specifications Document

Attach the final Design Specifications document you created. (Students insert a screenshot or export the page as an image here)

[Drag & drop or paste your Design Specifications image here]

Criteria B Strand ii – Annotated Design Ideas

Develop a range of feasible design ideas and annotate them to explain your decisions

DESIGN IDEA 1

[Name / Model 1]

[Drag & drop or paste screenshot of Idea 1 with annotations here]

[Briefly describe how this concept addresses your design specifications...]

DESIGN IDEA 2

[Name / Model 2]

[Drag & drop or paste screenshot of Idea 2 with annotations here]

[Briefly describe how this concept addresses your design specifications...]

DESIGN IDEA 3

[Name / Model 3]

[Drag & drop or paste screenshot of Idea 3 with annotations here]

[Briefly describe how this concept addresses your design specifications...]

Footer Checklist: ✔ Annotations are clearly visible • ✔ Three different creative concepts shown

Evaluating Design Ideas

Criteria B Strand iii – Selecting My Final Design

Critically evaluate your concepts against specifications to justify the chosen design path

My Final Choice

Selected Design: ☐ Idea 1 ☐ Idea 2 ☐ Idea 3

Justification: Explain why this design best meets your design specifications (4–5 sentences).

Type your comprehensive design justification here...

Criteria B Strand iv – Planning Diagrams

Create precise architectural diagrams and logical flows to systematically guide your build

System Planning Workbook

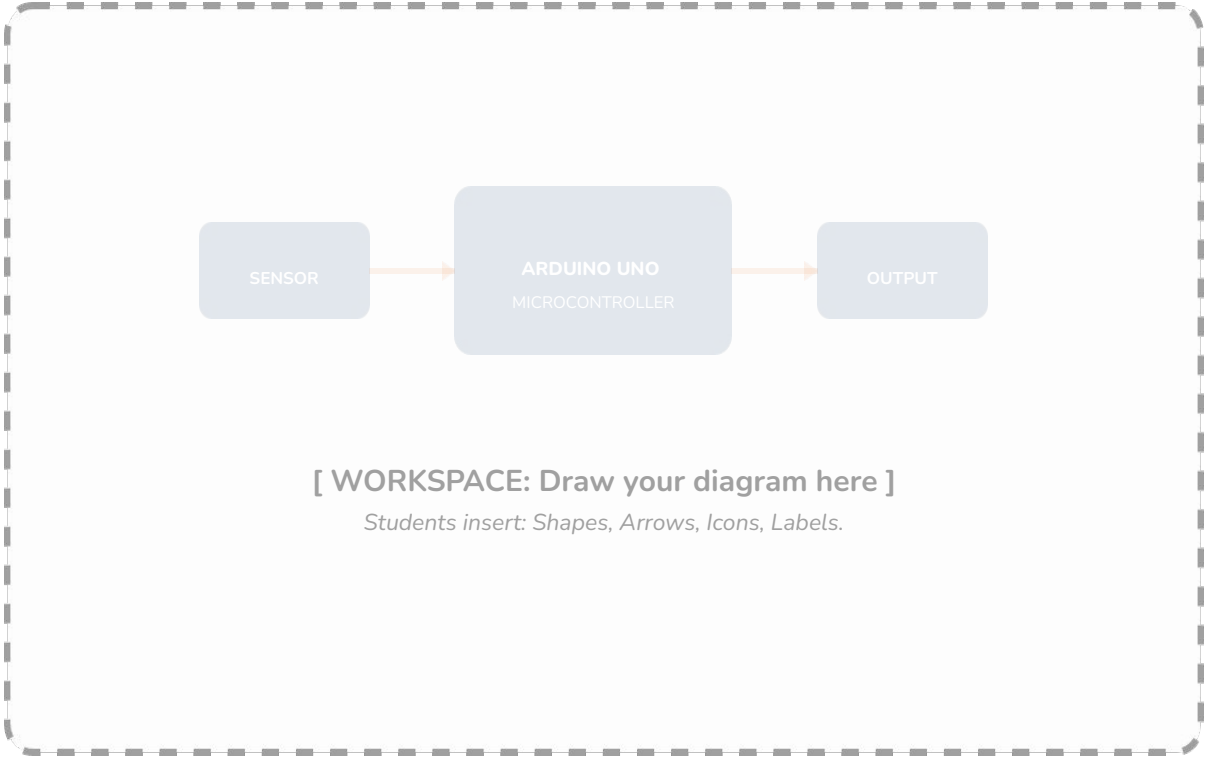
Planning Diagram – System Architecture

Components Checklist

[ ] Sensor(s)
[ ] Arduino Uno
[ ] Output Device(s)
[ ] User
[ ] Information Flow (Arrows)

DESIGN TIPS

- Use rectangles for components.
- Use arrows to show the direction of info.
- Label every component clearly.
- Keep your diagram neat and easy to read.



System Planning Workbook

Reflection & Self-Assessment

Part A: Looking Back (Answer Yes or No)

• My System Architecture clearly communicates my design.

[Type Yes/No here...]

• My Flowchart shows the complete system logic.

[Type Yes/No here...]

• I understand how my system will work before building it.

[Type Yes/No here...]

Part B: Reflection Questions

1. Which part of your planning are you most confident about?

[Type your answer here...]

2. What challenges do you expect when building your system in Wokwi?

[Type your answer here...]

3. What is one improvement you may need to make while creating your prototype?

[Type your answer here...]

Engineering Tip:

Reviewing your planning logic now saves troubleshooting hours later in Wokwi. Double-check your flowchart paths!



Save your work before submitting.